

UNISA STEM – SA PATHOLOGY

Laboratory Automation Solution

*A laboratory system with Collaborative Robot and Autonomous Mobile
Robot integration*

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1. Introduction

1.1. Background:

Over the past decade, robotic technologies in industrial applications have paved the way for a transition from traditional systems to automated ones, enhancing efficacy and precision while maintaining the availability and cost-effectiveness of the workflow. In the health sector, particularly in the laboratory environment, the demand for robotics and automatic systems has increased significantly due to the high volume of pathological testing and a shortage of human resources, intensified by the quarantine restrictions during the outbreak of the pandemic crisis 2020. To relieve the immense strain and pressure on the healthcare system, it is essential to apply automation solutions to improve and optimise the operations of the laboratory workspace.

1.2. Objective

The objective of the solution is to implement a human-robot collaboration (HRC) workspace within the laboratory environment that increases the proficiency of the system operation. By integrating cutting-edge robotic technologies like collaborative robots (cobots) and autonomous mobile robots (AMRs), the project provides an approach to enhance efficiency, accuracy and reduce reliance on human intervention in the pathological workflow, while upholding laboratory safety, mitigating error risks and ensuring human-like operation and augment productivity within the laboratory setting.

1.3. Scope:

- The scope of the solution is to increase the efficiency of the pathological procedure by confining the integration of cobots and AMRs in a laboratory system to automate the process of sample handling, processing and transportation within a reliable human-robot workspace.
- This will reduce human errors, improve traceability, and time efficiency and ensure performance consistency.
- The cognitive component in AMRs and cobots needs to be equipped the high-resolution imaging and analysis, and dynamic routing and navigating systems that are up-to-date technologies to ensure the quality of the system procedure.
- The requirements of regulatory safety and training compliance have to be met to maintain a safe working environment for human staff and ensure the continuous improvement of the operation.
- The integration of the automated system needs to ensure the comparability and coordination of the current system in data management and interoperability within the existing Laboratory Information System (LIS).
- The solution is flexible, integrable and available to be applied in multiple facility settings, however, in this proposal, we will focus on the solution in Royal Adelaide Hospital as the use case of the solution. ([Appendix 1](#))

2. Current Laboratory Workflow Analysis

2.1.Process mapping

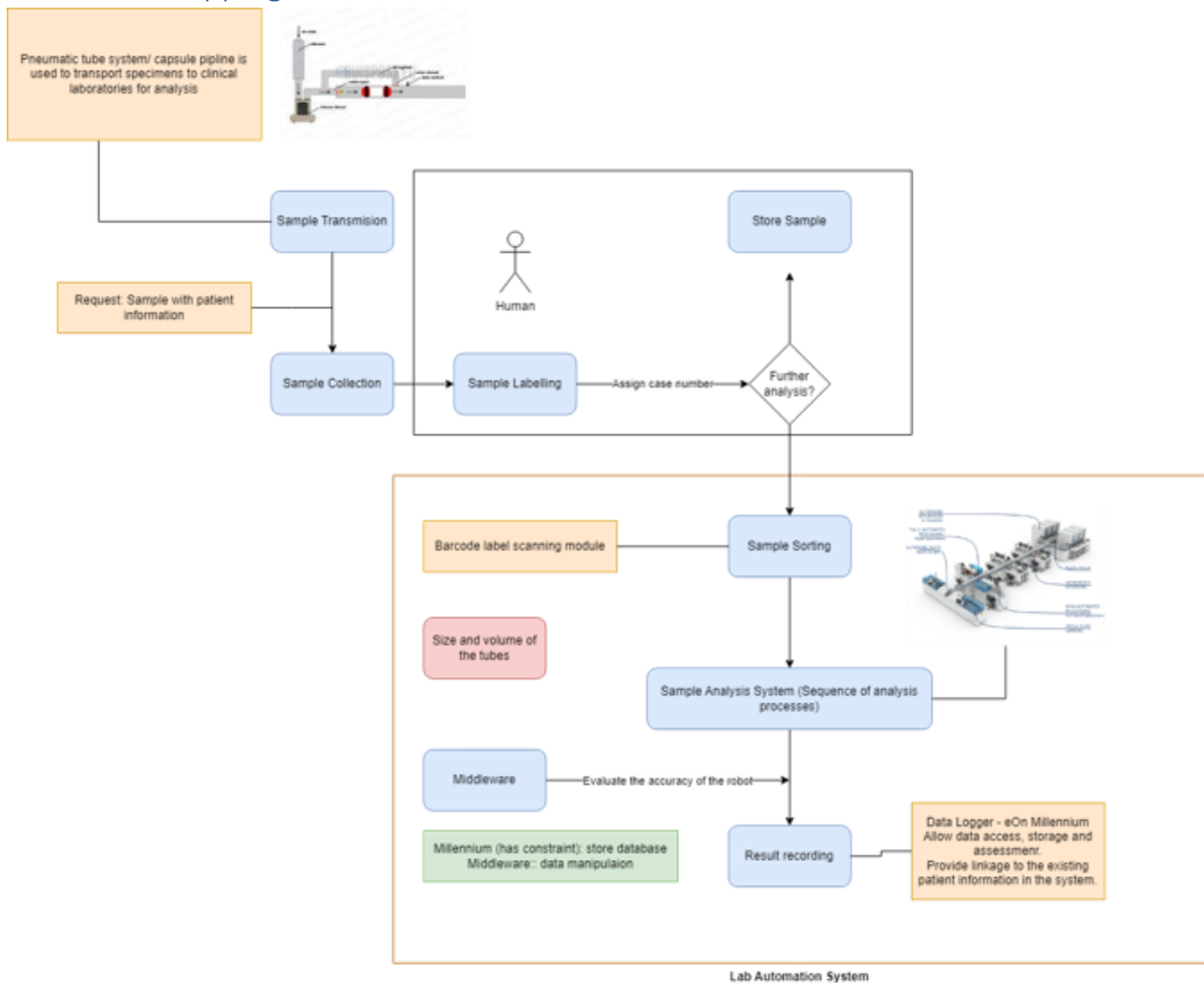


Figure 1. Current laboratory workflow (RAH)

Process phase	Actor(s)	Description
Sample Collection	Hospital staff (human) Pneumatic tube system	The specimens with sample information and details from patients collected in other departments are transported to the pathology lab via a pneumatic tube system.
Sample Logging	Pneumatic tube system	The samples are received in the capsule via the pipeline.
	Lab technicians (human)	Lab technicians check, enter and register the information of the patients and their samples into LIS.
	LIS	The system returns a label for each registered sample.
Sample Preparation	Lab Technicians (human)	Lab technicians prepare samples including centrifugation, pipetting and slide preparation.
Sample Scanning	Total Automation Solution	Lab technicians bring samples to the total automation label scanning machine.
	Lab Technicians (human)	The labels are scanned in the total automation solution system to identify which tests need to be run.
		Total Automation Solution assign samples in waiting for the pathological analysis.

Total Automation Solution (Sample Analysis)	Total Automation Solution	Total Automation Solution process the analytical testing based on the given requests from the label. Analyse, interpret and store the results and reports in the LIS.
Data management and manipulation (LIS and Middleware)	Data management software / medical / laboratory Millennium Helix™ Corner	Allow users to access and monitor the data results from the analysis.
Sample Storage and Disposal	Lab Technicians (human)	Lab technicians either store or dispose of the sample depending on the purposes of the analysis.

2.2.Risk identification and resource utilization

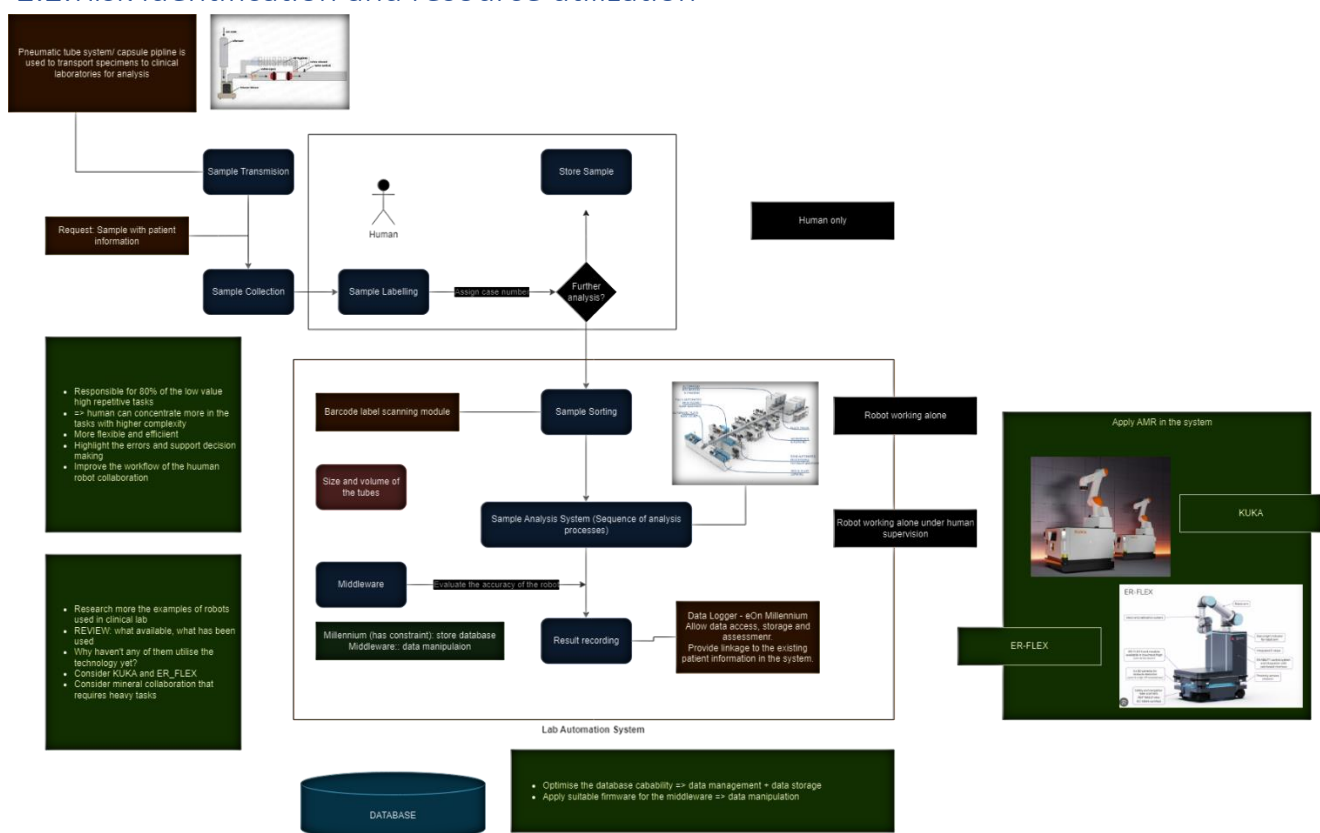


Figure 2. System Analysis and Recommendation

Risks and issues of the current system

According to the analysis of the current system, Sample Collection, Sample Logging, Sample Preparation and Sample Storage processes are monitored by humans, whose performance quality might be affected by the risk of human errors and system inefficiency.

Manual processes done by humans are time-consuming and labour-intensive, decreasing consistency and accuracy of the performance, which overall will affect the quality of the workflow, slow down the turnaround time and limit the availability of the system.

Human resource optimisation

Low value – high repetitive tasks like sample preparation, transportation and storing should not require manual labour, yet, could have been performed by robots working independently, under human supervision or within a human-robot collaboration environment.

With the current system, lab technicians are required to work and waste time on simple repetitive tasks instead of producing more value by spending time on tasks with higher value and complexity.

3. Requirements Specification

3.1.Functional Requirements

The manual processing executed by pure humans will need to be transferred to the automation workflow utilising cobots and AMRs in the laboratory setting under safety considerations and regulations. In detail, this solution provides three approaches to robot integration that will be applied in this laboratory system.

Robot setting	Target technology	Applied process	Description
Robot working independently	AMRs	Sample storage and disposal	AMRs will be notified by the system to keep track of the sample-handling process's real-time status. These robots will be isolated to perform tasks on their own within the set-up environment.
Robot working under human supervision	AMRs	Sample transportation	AMRs will be used to perform repetitive tasks in sample transportation within the system operation. AMRs will be used to deliver samples between different laboratory stations. These robots will work in the same area as the human workspace under restriction compliance.
Robot working with humans (human-robot collaboration)	Cobots	Sample preparation Sample assessment	Cobots will be used to prepare and assess samples with higher precision and consistency in system performance with less of risks from human errors. Cobots will be set at an isolated station and work alongside humans in the same workspace set-up.

The data integration between robots (AMRs and cobots) and existing LIS need to be established efficiently to maintain data integrity and accessibility throughout the system.

3.2. Non-Functional Requirements

- The solution should ensure high reliability and fault tolerance with an uptime of greater than 99% while minimising downtime with a highly efficient fault detection and recovery mechanism.
- The solution should provide good ease of use with user-friendly robot monitoring interfaces and comprehensive training and competency programs to ensure the continuous safety of the system.
- The solution should be scalable and expandable to adapt to the changes in the facility and the increasing volumes of pathological tests while maintaining the quality of the performance.
- The solution should comply with regulatory standards in Australia (according to NATA and TGA standards) with regular maintenance and monitoring documented.
- The solution should ensure the integrity and security of the data throughout the procedure and meet the quality assurance of the system.

4. System Design and Architecture

([Appendix 3](#) – 3D model version of the design)

Components	Integrated actor	Responsibility	Recommendation	Specification
Hardware	AMRs	Responsible for sample transportation and sample storage.	KMR iiwa KuKa AMRs (large area, heavy duty)	Max speed: 30.0 m/s Arm payload: Up to 14kg Mobile payload: Up to 200kg Operating time: depends on Lion battery Precision: +/- 0.1 mm Safe, stable and flexible. Equipped with a sensor and scanner.
			EF-250-UR5e ER-FLEX robot (smaller and lighter duty)	Max speed: 2.0 m/s Arm payload: Up to 16kg Mobile payload: Up to 186kg Operating time: Up to 20 hours/day (4.5 hrs/charge) Precision: within 1mm Easy to integrate into the existing storage system. Safe, stable and flexible. Equipped with a sensor and scanner.
	Cobots	Responsible for sample handling and preparation.	YuMi Single-arm cobot	Reach: 0.559 m Arm payload: 0.50 kg Mounting: any angle Joints: 7 Easy to integrate into the existing storage system. Safe, stable and flexible. High precision and fast performance. Mounting in any position Integrated gripper with vision and vacuum Best-in-class safety design High speed and accuracy
		Responsible for human assistance in contagious or high-accuracy tasks.		
Software	AI-based image analysis software	Responsible for digital image processing and dynamic environment navigation and mapping. Responsible for sample handling process that requires sample classification	Object Detection model (Computer vision YOLOv8) SLAM (simultaneous localization and mapping) RBGD & LiDAR (depth and colour camera with light detection and ranging)	Both AMRs and Cobots will be integrated with the latest technology of computer vision and image processing with the high accuracy navigation and sensor system.
	Middleware	Responsible for data management and manipulation.	Orion Health Rhapsody Integration Engine	Supports a wide range of healthcare messaging standards, including HL7, FHIR, and DICOM.

		Facilitates seamless data integration across various healthcare systems		Flexible to integrate into both legacy systems and modern applications. Comprehensive security features, including encryption, user authentication, and audit trails. Be able to scale with the increasing data exchange demands of growing laboratories. Complied with Australian health data standards and privacy regulations
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5. Implementation Plan

Workflow in system design solution ([Appendix 2](#))

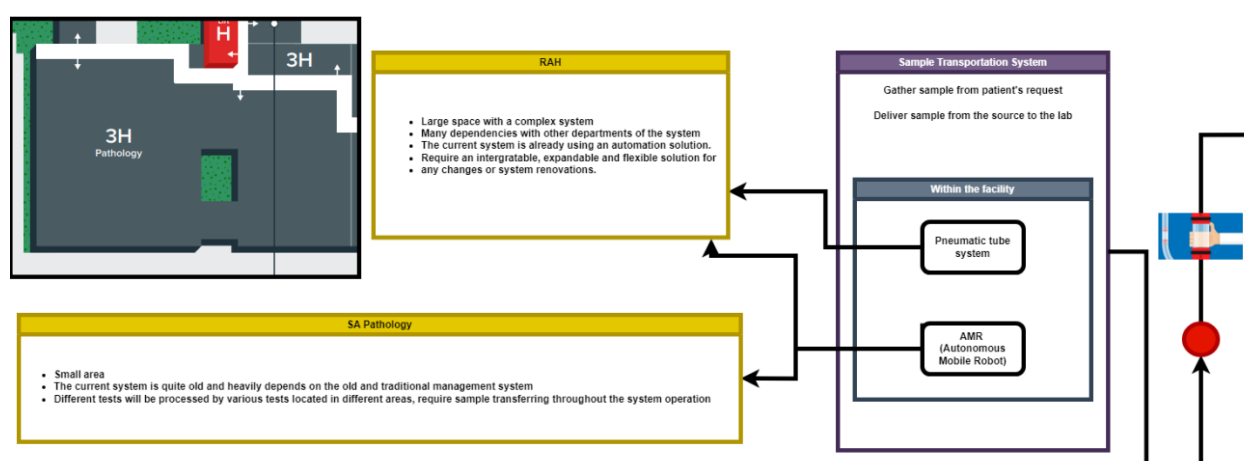
Sample Transportation (before arrival Pathology) system

The pathological test requests from patients are collected and sent through the sample transportation system where:

- RAH:

RAH is a large area and equipped with up-to-date facility will remain using the pneumatic tube system for the implementation of the sample delivery.
- SA Pathology:

SA Pathology with smaller facility and is not equipped total automation solution and pneumatic tube in the system due to the limitation of the area integrate better with the utilisation of AMR in system delivery.



Pre-analytical phase

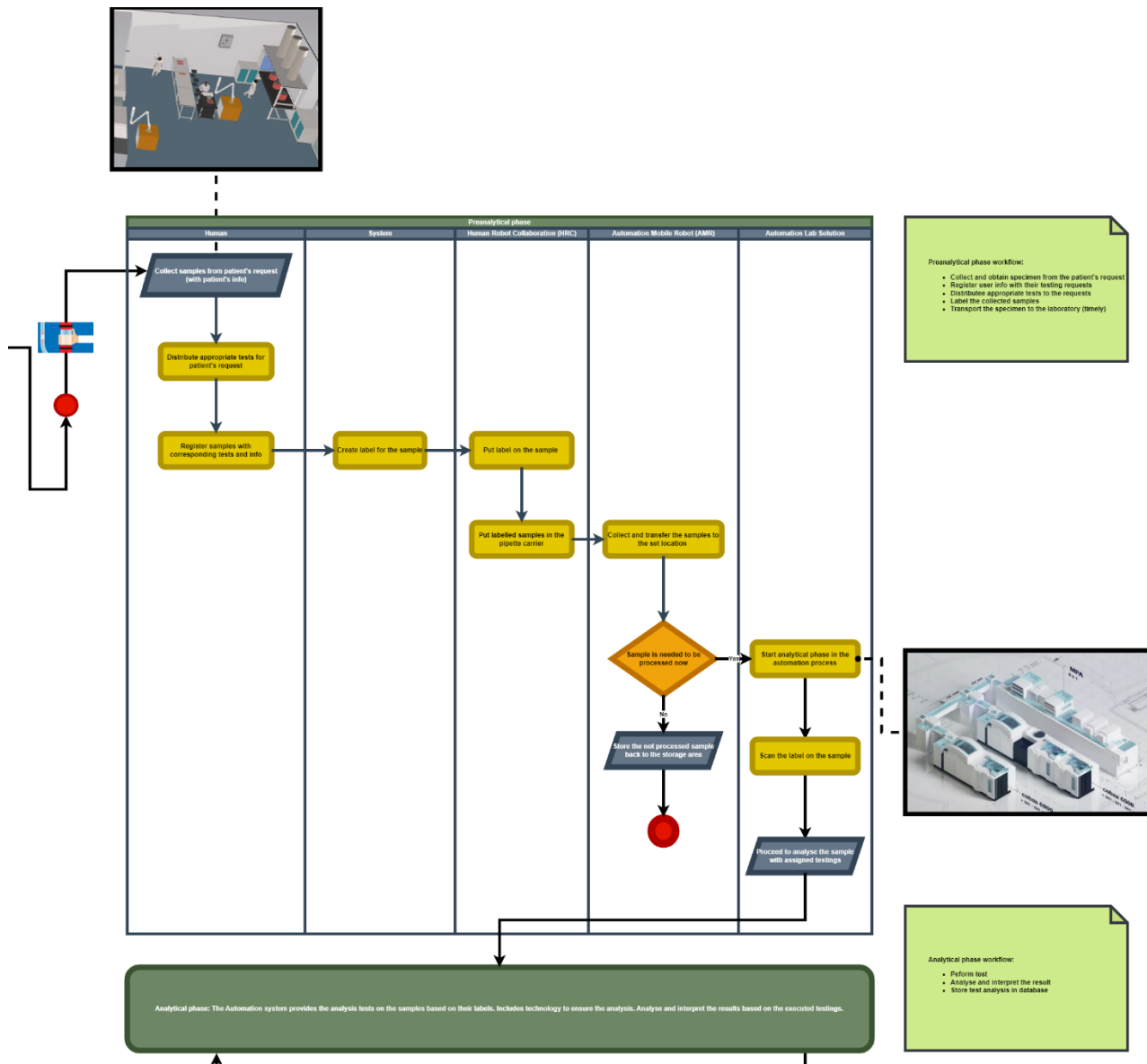
The lab technicians work with cobots in sample labelling after receiving sample capsules via the transportation system.

- The lab technicians enter sample information and patient details into the system.

- The system stores the data and generates a scanning label uniquely for each sample.
- Cobot scans, analyses and assists humans in stacking the sample on the rack holder.
- Cobot handles the sample preparation process with humans during this step.

UNDER WORKING REGULATION:

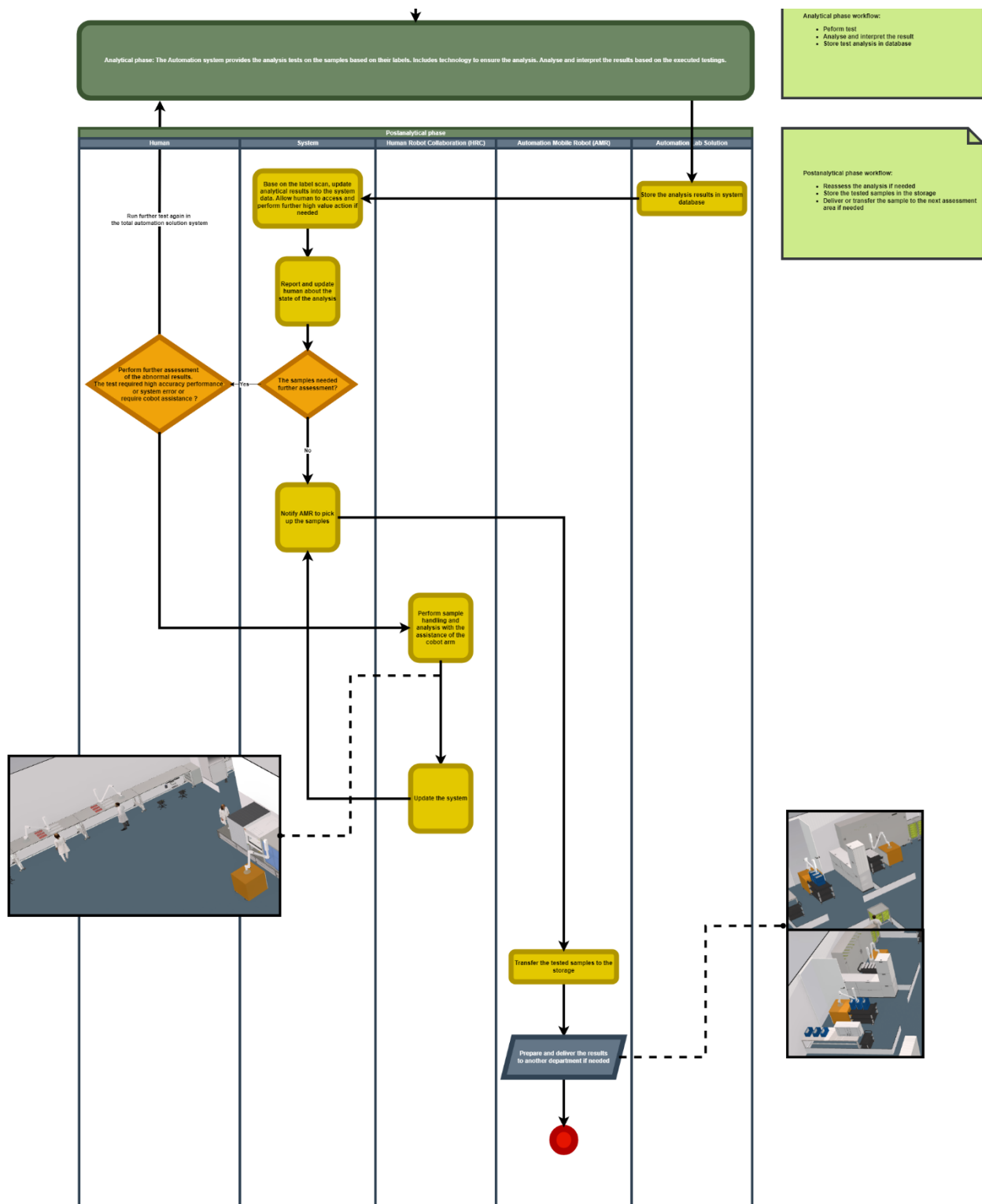
AMR collects samples from the rack and transfers them to the start of the total automation system. If the samples are not meant to be processed now, AMR will transfer them to the awaiting storage area.



Analytical phase

The total automation solution system has the same responsibility and functions that execute the flow of pathological tests.

The proposed middleware is applied in this phase to handle and monitor the performance of the automation solution, the LIS and the data storage to ensure the accuracy and quality of the system and data management performance.



Post-analytical phase

The system finishes the analysis phase and notifies the human about the results and abnormal analytical results.

- The system will notify the AMR to pick up the samples to dispose of or store them back in the storage if there are no further actions needed to be done.
- Pathologists decide whether the test needs to be re-run or not and whether to re-execute the test via a total automation system again or use cobots to perform further analysis.
- After finished, the human-robot collaboration setting reports to the system, therefore, the system notifies the AMR to bring it back to either storage or disposal area.

APPENDIX 1



Figure 3. RAH Pathology area - floor plan level 3

APPENDIX 2

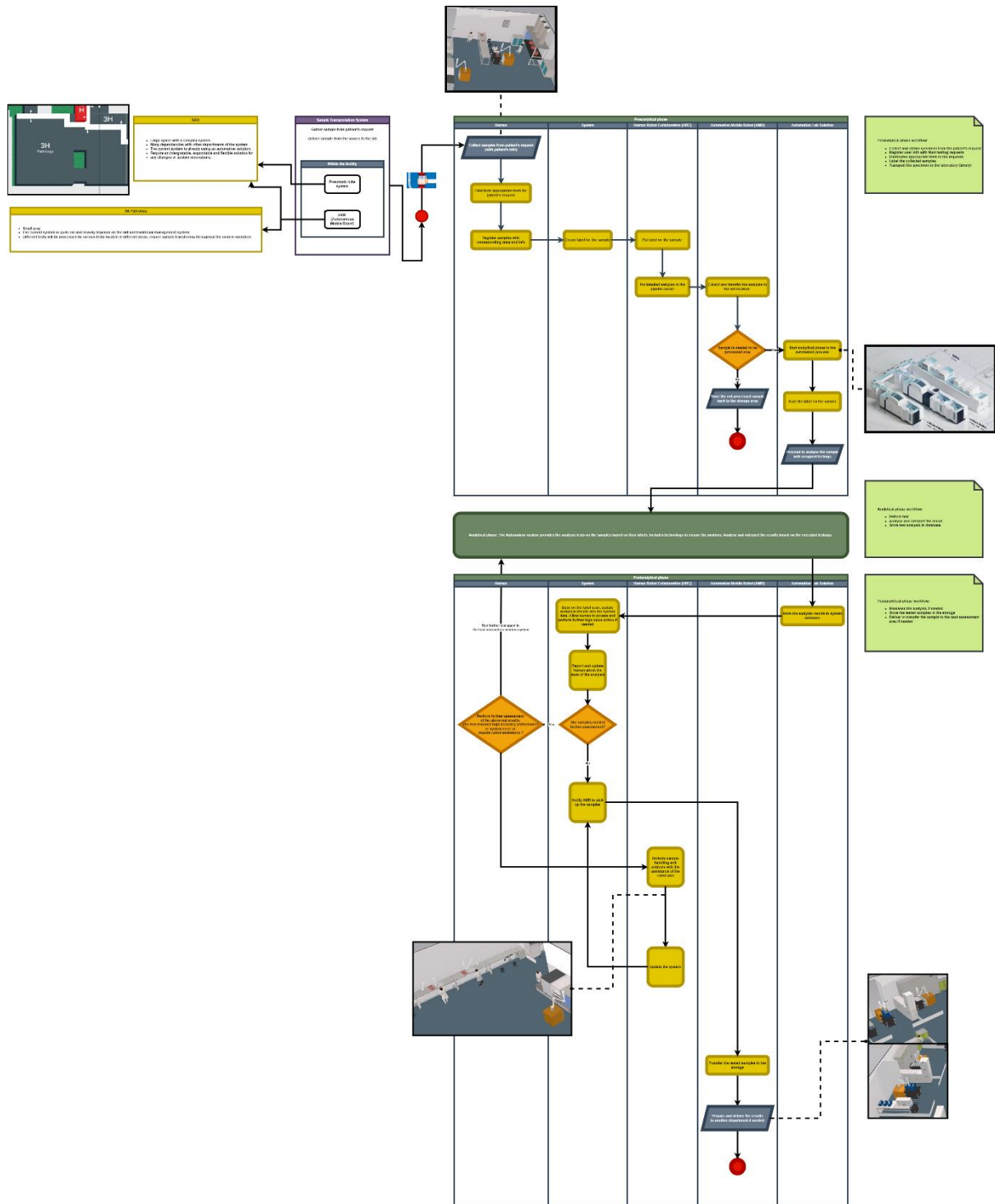


Figure 4. System Design -workflow in RAH

APPENDIX 3

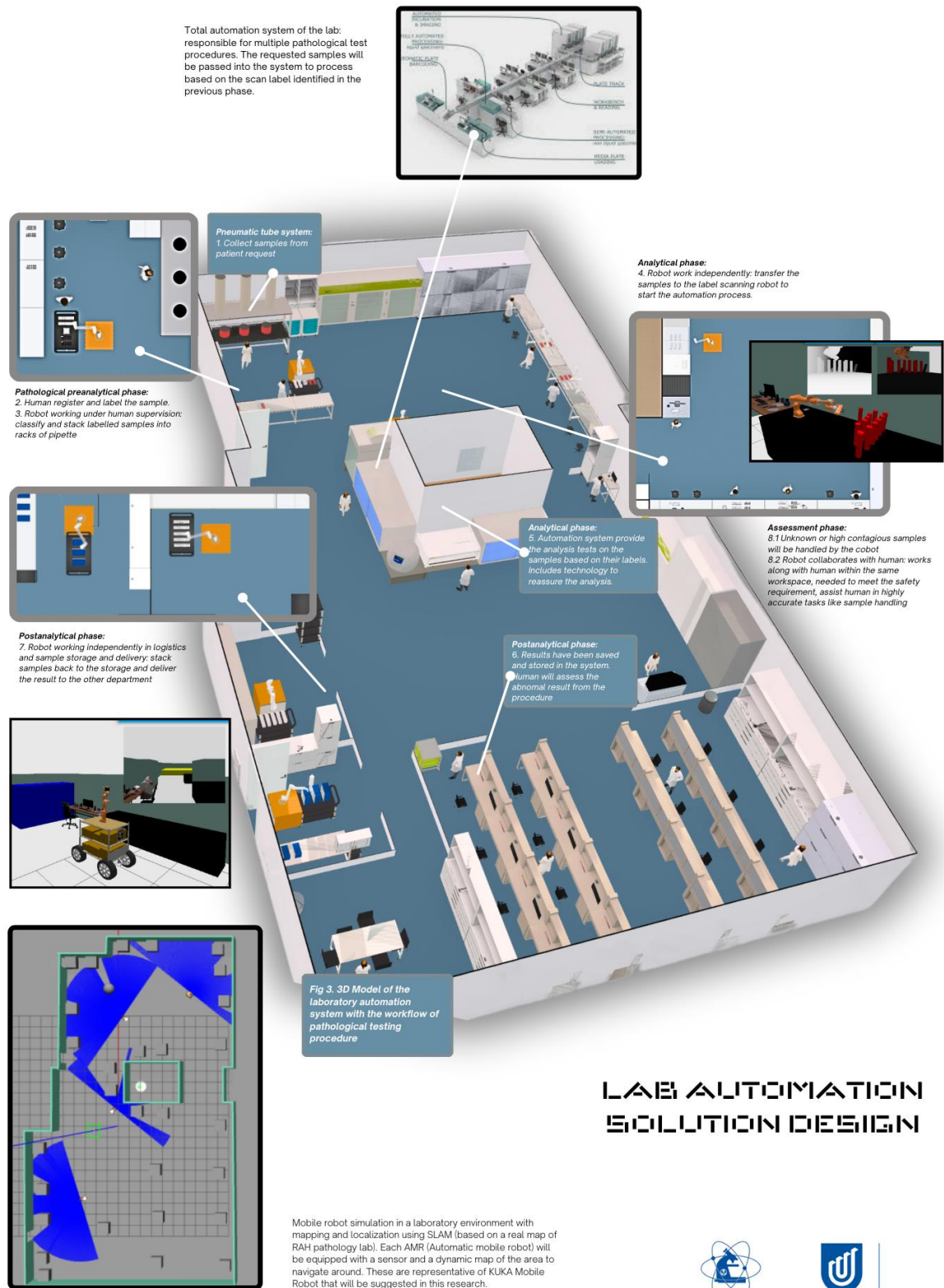


Figure 5. Lab automation solution design

APPENDIX 4

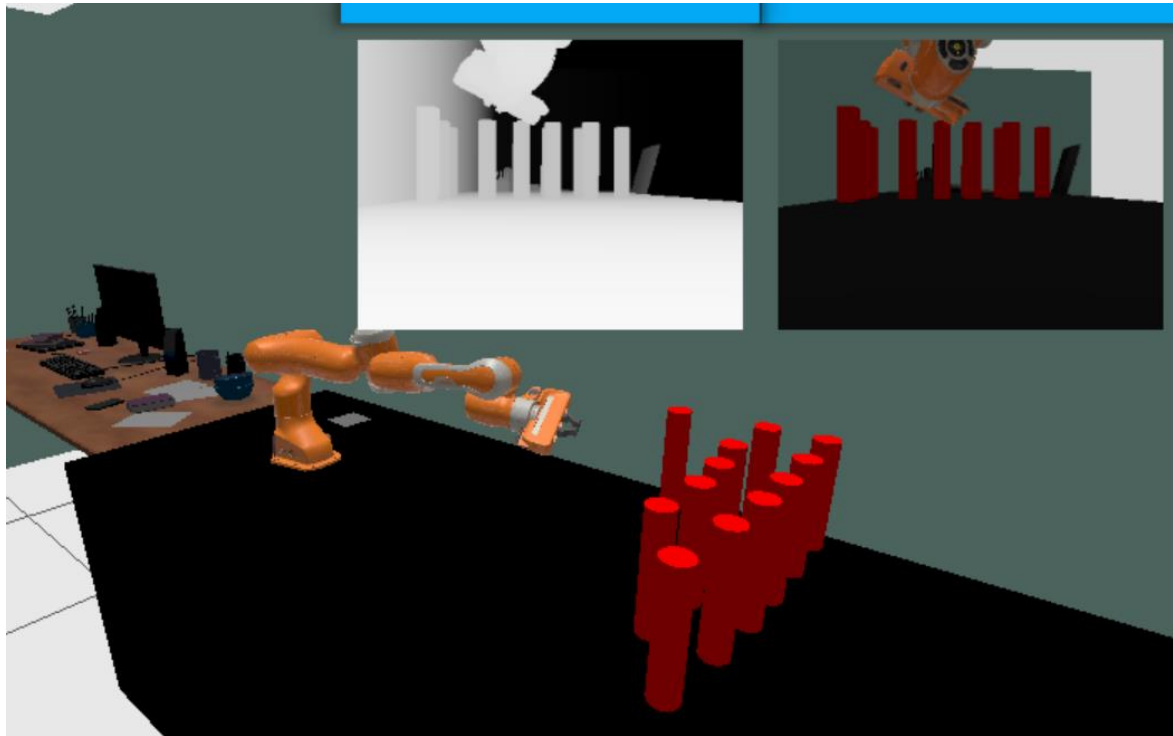


Figure 6. Robot Simulation in Gazebo - Cobot arm



Figure 7. Robot simulation in Gazebo - AMR